# FORENSIGHT • CYBERSECURITY INCIDENT INVESTIGATION REPORT

Digital Forensics & Incident Response (DFIR) Dossier • Generated: 2026-10-20 15:51:44 UTC

**TLP:AMBER | STRICTLY CONFIDENTIAL — FOR AUTHORIZED INVESTIGATIVE & LEGAL USE ONLY**
This document contains proprietary forensic technical analysis, evidence hashes, and attack trajectory models. Unauthorized reproduction, export, or distribution outside designated incident response personnel is strictly prohibited.

## 1. Executive Incident Summary

This formal incident investigation dossier documents the digital forensic examination and analytical findings associated with ForenSight Case #19, titled "Operation Nightshade: APT39 Incursion". The investigation was initiated on 2026-10-19 19:23:06 under the authority of Lead Investigator Sarah Connor and is classified as a Critical-priority Advanced Persistent Threat incident. Current operational disposition is Open.

The investigative trigger originated from initial recorded alerts indicating: "Evidence of lateral movement and pass-the-hash attacks detected on DC-01.". To determine the scope, validity, and impact of the suspected adversary activity, the DFIR team secured 1 digital evidence artifact(s) under strict ISO/IEC 27037 chain-of-custody protocols. Subsequent laboratory analysis has cataloged 1 confirmed technical finding(s) (1 Critical, 0 High, 0 Medium, 0 Low).

Forensic examination confirms active threat indicators consistent with advanced persistent threat tradecraft. However, in accordance with scientific forensic standards, this report strictly delineates between verified host-level observations, investigator-recorded working hypotheses, and unverified assumptions. Immediate containment actions and secondary forensic procedures are detailed within this document.

| | | | |
|---|---|---|---|
| **Case Identifier** | FS-19 | **Incident Classification** | Advanced Persistent Threat |
| **Case Title** | Operation Nightshade: APT39 Incursion | **Priority Level** | **Critical** |
| **Investigation Status** | Open | **Lead Investigator** | Sarah Connor (Investigator) |
| **Incident Opened** | 2026-10-19 19:23:06 | **Investigator Email** | test_investigator_e2e@forensight.io |
| **Seized Evidence Files** | 1 artifact(s) (SHA-256 verified) | **Identified Threat Findings** | 1 finding(s) (1 Critical) |

## 2. Incident Background and Operational Scope

Investigation FS-19 was formalized following anomalous telemetry flagged within the organization's operating environment. The scope of this inquiry is defined by the narrative parameters: "Evidence of lateral movement and pass-the-hash attacks detected on DC-01.". The primary focus centers on identifying the mechanisms, timeline, and objectives of the suspected adversary, specifically evaluating whether privileged domain assets were successfully compromised.

Operational context: The target asset identified in the initial reporting (DC-01) functions as a mission-critical Active Directory Domain Controller. In enterprise infrastructure, Domain Controllers maintain the central identity vault (NTDS.dit), coordinate Kerberos Ticket Granting Services (TGS), and enforce domain-wide Group Policy Objects (GPOs). Compromise of a Domain Controller represents a Tier-0 security event capable of yielding total administrative control over the entire enterprise forest.

Investigation Boundaries: This report assesses the physical and logical artifacts currently deposited in the ForenSight evidence vault. It explicitly does not evaluate unacquired endpoints or third-party cloud services unless corroborating records have been ingested into the custody log.

## 3. Incident Classification and Severity Rationale

Incident Classification: Advanced Persistent Threat. Priority Assignment: Critical.

Advanced Persistent Threat (APT) classifications are designated for intrusions demonstrating sophisticated, evasive tradecraft typically associated with organized state-sponsored cyber espionage units or highly resourced threat groups. Key characteristics include deliberate living-off-the-land techniques (LotL), memory-only payload execution, process injection into benign system processes, long-term dwell time, and stealthy lateral movement.

Severity Rationale (Critical): Under the ForenSight Incident Triage Matrix, a Critical designation is justified by the criticality of the targeted infrastructure, the presence of 1 critical/high findings, and the potential blast radius. If active credential theft or lateral movement is confirmed on a Tier-0 asset, enterprise-wide identity compromise becomes an imminent risk.

## 4. Initial Detection Telemetry and Indicators

Initial detection was logged on 2026-10-19 19:23:06. The opening event is captured in the case record: "Evidence of lateral movement and pass-the-hash attacks detected on DC-01.". This record outlines the initial observations that warranted forensic escalation.

Pass-the-Hash (PtH) & Lateral Movement Indicators: Initial triage flagged anomalies characteristic of Pass-the-Hash attacks (MITRE ATT&CK; T1550.002). Pass-the-Hash allows adversaries who possess an NTLM password hash to authenticate to remote network services (such as SMB over TCP port 445 or WMI/RPC over TCP port 135) without ever knowing or cracking the plaintext password. This technique is standard tradecraft for pivoting between workstations and domain servers.

Secondary Detection — Finding #9 (Process Injection): Recorded observation on apt39_memdump.raw: "Reflective DLL injection detected inside lsass.exe process address space.".

## 5. Investigation Objectives and Charter

1. Evidence Authenticity & Cryptographic Verification: Validate SHA-256 integrity checksums across all submitted artifacts to ensure evidentiary admissibility under ISO/IEC 27037 standards.

2. Scope & Target Triage: Determine whether adversary activity was contained or successfully compromised DC-01.

3. In-Memory Process Analysis: Conduct forensic examination of volatile memory artifacts to detect unlinked dynamic link libraries (DLLs), hollowed processes, and injected shellcode buffers.

4. Credential Access Validation: Determine whether the Local Security Authority Subsystem Service (lsass.exe) was accessed, dumped, or modified by injected code.

5. Lateral Movement Verification: Cross-examine network states and host event logs to establish whether reported pass-the-hash attempts were blocked or succeeded.

6. Threat Nexus & Attribution Scrutiny: Evaluate investigator notes regarding malware family or threat group associations (e.g., APT39 implant variants) against empirical forensic standards.

7. Remediation Directives: Formulate prioritized containment, eradication, and post-incident hardening recommendations.

## 6. Evidence Acquisition and Integrity Verification

Evidence ingestion was conducted under strict forensic standards. A total of 1 digital artifact(s) were secured and indexed into the ForenSight vault. Cryptographic SHA-256 hashes were calculated immediately upon ingestion to establish an immutable baseline.

• Volatile Memory Dump: Artifact #17 ("apt39_memdump.raw", 0.1 KB, SHA-256: f1eafcc022fea7e9...). Volatile memory (RAM) is the most critical and fragile evidentiary source in modern cyber investigations. Because memory-only payloads, decrypted network streams, unbacked injected code, and active logon sessions disappear upon system power-off or reboot, volatile memory acquisition takes precedence under the Order of Volatility (RFC 3227).

Chain of Custody Enforcement: A total of 2 custodial event(s) have been logged. All digital evidence items remain stored in write-protected, restricted-access storage. Every transfer, integrity hash verification, and analyst access is recorded with timestamps and responsible investigator identity.

## 7. Investigation Methodology and Frameworks

The DFIR engagement adheres to the NIST SP 800-61 Rev. 2 Computer Security Incident Handling Guide, complemented by the MITRE ATT&CK; Matrix for Enterprise and ISO/IEC 27037 standards for digital evidence handling:

**Phase 1: Evidence Triage & Cryptographic Baseline**: All seized artifacts are matched against their initial SHA-256 acquisition hashes. Working copies are created for analytical tools to prevent modification of primary evidentiary images.

**Phase 2: Volatile Memory Forensics & Process Tree Reconstruction**: Memory dumps are inspected using framework plugins (such as Volatility 3) to enumerate running processes, map Virtual Address Descriptors (VAD), identify unlinked executable memory sections, and analyze security subsystem threads.

**Phase 3: Finding Correlation & Tradecraft Mapping**: Logged anomalies are correlated with known adversary Tactics, Techniques, and Procedures (TTPs) using the MITRE ATT&CK; framework. Observations are checked against known operating system baseline behaviors to filter benign false positives.

**Phase 4: Hypothesis Testing & Limitation Assessment**: The investigative team rigorously tests hypotheses against observable evidence. Unverified claims (e.g., successful credential theft or external command-and-control communication) are explicitly flagged as hypotheses requiring additional telemetry.

**Phase 5: Remediation Strategy Formulation**: Concrete containment and hardening directives are developed, prioritized by risk and operational urgency.

## 8. Detailed Technical Analysis and Findings Interpretation (1 records)

This section provides comprehensive technical analysis of each recorded finding. In accordance with forensic rigor, each observation is analyzed for its technical mechanics, forensic significance, what it establishes, and explicitly what it does NOT prove without secondary telemetry.

**Finding #9: Process Injection | Severity: Critical | Artifact: *apt39_memdump.raw***

**Recorded Observation:** Reflective DLL injection detected inside lsass.exe process address space.

**Confidence Level: High Analytical Confidence (Host Memory Observation)**

Technical Mechanics & Significance: The Local Security Authority Subsystem Service (lsass.exe) is a critical Windows operating system component responsible for enforcing security policies, authenticating users, generating access tokens, and managing active Kerberos, NTLM, and WDigest credential structures in memory. Reflective DLL injection is an evasive technique where a dynamic link library is injected into a running process's virtual address space directly from a memory buffer without calling the standard Windows LoadLibrary API or writing the binary to disk. This bypasses file-system write monitoring and prevents the DLL from being registered in the process's InLoadOrderModuleList within the Process Environment Block (PEB). Adversaries target lsass.exe specifically to execute in-memory credential harvesting tools (such as Mimikatz or custom memory scavengers) to dump plaintext passwords, Kerberos Ticket Granting Tickets (TGTs), and NTLM password hashes for subsequent Pass-the-Hash lateral movement.

**CONFIRMED OBSERVATION:** What This Finding Confirms: Confirms that an in-memory modification or anomalous executable memory region (exhibiting PAGE_EXECUTE_READWRITE or unbacked VAD characteristics) was detected within the virtual address space of lsass.exe on the analyzed host memory artifact.

**CRITICAL LIMITATION:** What This Finding Does NOT Prove: CRITICAL LIMITATION — This finding does NOT independently prove that credentials were successfully dumped, decrypted, or exfiltrated by the adversary. It does not establish whether the injected thread executed to completion, was terminated by an endpoint agent, or was intercepted. Furthermore, it does not prove that lateral movement beyond the host succeeded. These downstream impacts remain investigative hypotheses until corroborated by network traffic logs, secondary endpoint memory dumps, or Active Directory authentication logs.

**Proposed Validation Procedures & Command Syntax:**

**Proposed Command / Analysis Syntax:** Identify unmapped executable memory regions and injected code blocks:

```
python3 vol.py -f apt39_memdump.raw windows.malfind
```

**Proposed Command / Analysis Syntax:** Inspect process module lists to find unlinked or hidden DLLs not in PEB:

```
python3 vol.py -f apt39_memdump.raw windows.ldrmodules --pid
```

**Proposed Command / Analysis Syntax:** Verify the process tree and parent-child hierarchy of lsass.exe (expected parent: wininit.exe):

```
python3 vol.py -f apt39_memdump.raw windows.pstree
```

**Proposed Command / Analysis Syntax:** Dump suspicious VAD memory allocations to disk for string extraction and YARA rule matching:

```
python3 vol.py -f apt39_memdump.raw windows.vaddump --pid --dump
```

**Next Analytical Step:** Validation Protocol: Execute the proposed Volatility3 plugins against the primary memory dump. Inspect malfind output for MZ/PE headers or shellcode prologues (e.g., 0xFC, 0x48, 0x31) in unbacked memory pages. Cross-reference the injected code against the YARA repository for known credential dumping signatures.

## 8.1 Threat Intelligence Nexus and Field Notes Evaluation

Threat Intelligence Evaluation (Note #7 by Sarah Connor): "Cryptographic signature matches known APT39 implant variant (Shaheen-IV)."

Context & Actor Profile: APT39 (also tracked as Chafer, Remix Kitten, and ITG07) is a suspected Iranian state-sponsored cyber espionage collective active since at least 2014. The group has historically targeted telecommunications, travel, government, and IT service providers across the Middle East, Europe, and North America. Their operational tradecraft frequently employs custom backdoors (such as POWBAT, SEAWEED, and the Shaheen/Caterpillar family), alongside modified open-source tools (Mimikatz, RemCom, Nbtscan) for internal credential access and lateral movement.

CRITICAL FORENSIC BOUNDARY — ATTRIBUTION CAVEAT: The association of this memory artifact with an APT39 implant variant (Shaheen-IV) represents an investigative lead and working hypothesis recorded by the investigator. In professional forensic practice, tool-level similarities, shared code segments, or cryptographic matches DO NOT constitute definitive nation-state attribution. Threat actors frequently copy code, utilize leaked builder frameworks, or stage false-flag artifacts to complicate attribution. Attribution can only be established with high confidence when code similarity is corroborated by independent command-and-control infrastructure overlap, operational timing, and external threat intelligence confirmations.

## 9. Reconstructed Incident Timeline (6 confirmed milestones)

The following chronological matrix reconstructs the factual sequence of events established from immutable database audit records, custody transitions, evidence seizure, and detection timestamps:

| Timestamp (UTC) | Category | Milestone Event / Recorded Action | Actor / Origin |
|---|---|---|---|
| 2026-10-19 19:23:06 | Case Initiation | Investigation opened: Operation Nightshade: APT39 Incursion | Sarah Connor |
| 2026-10-19 19:23:06 | Evidence Seizure | Artifact #17 registered (apt39_memdump.raw) | Sarah Connor |
| 2026-10-19 19:23:06 | Custody Transfer | Evidence Ingested (apt39_memdump.raw) | Sarah Connor |
| 2026-10-19 19:23:07 | Custody Transfer | Transferred to Secure Vault (apt39_memdump.raw) | Sarah Connor |
| 2026-10-19 19:23:07 | Threat Detection | [Critical] Process Injection on apt39_memdump.raw | Forensics Engine |
| 2026-10-19 19:23:07 | Workflow Task | Task #8: Analyze Memory Dump for Injected DLLs (Pending) | Malishka Sachdev, Sarah Connor |

## 10. Suspected Attack Progression and MITRE ATT&CK; Matrix Alignment

The hypothesized attack chain is mapped against the MITRE ATT&CK; framework below. Each stage explicitly distinguishes between confirmed evidentiary artifacts and unverified investigative hypotheses:

| Kill Chain Phase | Tactic & Technique | Evidentiary Status | Analytical Narrative & Telemetry |
|---|---|---|---|
| 1. Initial Access | **TA0001: Initial Access** T1566: Phishing / T1078: Valid Accounts | **Investigative Hypothesis / Telemetry Missing** | Initial entry vector remains unverified in current records. Potential vectors include phishing delivery, perimeter credential abuse, or unpatched external services. Additional firewall and VPN authentication logs are required to establish the initial breach point. |
| 2. Execution & Evasion | **TA0002: Execution / TA0005: Defense Evasion** T1055.001: Process Injection: Dynamic-link Library Injection | **Confirmed Host Observation (Evidence #17)** | Confirmed by laboratory finding: Reflective DLL injection detected within the memory space of lsass.exe on DC-01. Bypasses disk-based security controls by executing purely in volatile virtual memory. |
| 3. Credential Access | **TA0006: Credential Access** T1003.001: OS Credential Dumping: LSASS Memory | **Probable Intent / Execution Unverified** | The targeting of lsass.exe indicates a high probability of intent to harvest cached NTLM hashes, Kerberos tickets, or LSA secrets. However, successful extraction is not yet confirmed by the current evidence record. |
| 4. Lateral Movement | **TA0008: Lateral Movement** T1550.002: Use Alternate Authentication Material: Pass the Hash | **Reported Initial Indicator / Pending Network Log Corroboration** | Case briefing notes report lateral movement and Pass-the-Hash targeting DC-01. Verification requires ingesting Windows Security Event Logs (Event IDs 4624 Logon Type 3 with NTLM package) and SMB session logs. |

| | | | |
|---|---|---|---|
| 5. Command & Control / Persistence | **TA0011: Command and Control / TA0003: Persistence** T1071: Application Layer Protocol / T1543: Create or Modify System Process | **Investigative Hypothesis (Shaheen-IV Implant Variant)** | Investigator notes associate the memory artifact with an APT39 implant variant. C2 beaconing and persistence hooks remain unverified until memory dump extraction and network egress matching are performed. |

## 11. Impact and Risk Assessment

Confidentiality Risk (HIGH TO CRITICAL): If the adversary successfully extracted credentials from lsass.exe on DC-01, the confidentiality of domain administrative accounts, service principal keys, and Active Directory Kerberos golden ticket signing keys (krbtgt) is severely compromised. Exposure of domain credentials permits unrestricted access to enterprise email, internal file repositories, and sensitive databases.

Integrity Risk (HIGH): Active process injection on a Domain Controller grants the attacker arbitrary kernel/SYSTEM-level execution rights. Risks include modification of Active Directory group memberships (e.g., Domain Admins), alteration of Group Policy Objects (GPOs), and tampering with security audit log configurations.

Availability Risk (MODERATE): To date, no disruptive payload (such as disk wiping or ransomware encryption) has been detected. However, operational availability may be impacted during emergency isolation, forensic triage, and mass credential revocation across the domain infrastructure.

Regulatory & Compliance Exposure: The potential compromise of enterprise identity credentials triggers reporting evaluations under regulatory frameworks (e.g., GDPR Article 33, HIPAA, PCI-DSS). If personal data or cardholder environments are accessible via compromised credentials, formal regulatory disclosure obligations may apply.

## 12. Evidence Gaps and Investigative Limitations

A complete and legally defensible forensic report must transparently identify what information is missing. The following gaps represent telemetry sources not currently linked to this case record and detail the exact validation protocols required to resolve them:

| Identified Telemetry Gap | Forensic Consequence & Impact | Required Remediation & Validation Protocol |
|---|---|---|
| 1. Absence of Network Packet Captures (PCAP) & NetFlow Data | The current evidence catalog contains no network packet captures or router flow records. As a consequence, outbound command-and-control (C2) beaconing, data exfiltration volume, and remote attacker IP addresses cannot be empirically verified. | Acquire perimeter firewall connection logs, DNS resolver query logs, and NetFlow records for DC-01 covering 72 hours prior to and following the initial detection timestamp. |
| 2. Missing Windows Security Event Log Telemetry (EVTX) | The initial briefing notes Pass-the-Hash and lateral movement targeting DC-01, but the raw Security Event Log (Security.evtx) has not been ingested. Without event logs, authentication success cannot be distinguished from failed attempts. | Export and ingest Security.evtx from DC-01. Filter for Event ID 4624 (Logon Type 3: Network, Package: NTLM, Key Length: 0), Event ID 4672 (Special Privileges Assigned), and Event ID 4720 (Account Created). |
| 3. Incomplete Volatility Analysis Execution (Task #8 Pending) | Investigation Task #8 ("Analyze Memory Dump for Injected DLLs") is currently marked as "Pending". The automated output of Volatility 3 malfind and ldrmodules plugins has not yet been committed to the database. | Execute the proposed Volatility 3 command sequence against apt39_memdump.raw and upload the parsed findings into the ForenSight AnalysisFindings table. |
| 4. Attribution Verification Boundaries | Investigator Note #7 cites an APT39 implant variant (Shaheen-IV). However, no YARA scan matches, compile timestamps, or C2 IP infrastructure overlaps have been formally verified in the laboratory record. | Extract the injected memory segment via vaddump, calculate its ssdeep fuzzy hash, and execute YARA rules specific to APT39 Chafer implants. Cross-reference results with external threat intelligence feeds. |

## 13. Forensic Conclusions and Case Disposition

Investigation FS-19 establishes with high forensic confidence that an in-memory threat indicator (Reflective DLL injection into lsass.exe) is present on the submitted memory artifact from DC-01. This tradecraft strongly aligns with credential harvesting and lateral movement methodologies typical of advanced threat actors.

CRITICAL EVIDENTIARY STATUS: The evidence currently available in the case repository establishes active host-level targeting, but DOES NOT confirm successful credential exfiltration, full domain compromise, or definitive nation-state attribution to APT39. These possibilities remain working hypotheses that require the ingestion of network traffic logs, Windows event logs, and completed memory dissection results.

Case #19 remains in OPEN / ACTIVE INVESTIGATION status. The physical and logical integrity of all seized artifacts is fully preserved under ISO/IEC 27037 standards with verified SHA-256 cryptographic hashes.

## 14. Prioritized Containment, Eradication, and Recovery Directives

The following actionable directives are organized by operational phase to guide immediate containment and long-term structural eradication:

### Phase 0: Immediate Containment (Hours 0 – 4) [CRITICAL / P0]

1. Network Isolation: Logically isolate DC-01 from all non-essential internal subnets using host firewall rules or switch port ACLs, while maintaining power to avoid destroying volatile memory contents.

2. Privileged Credential Revocation: Enforce an immediate password reset and Kerberos ticket invalidation for all Domain Administrator, Enterprise Administrator, and privileged service accounts from an uncompromised administrative workstation.

3. Invalidate Kerberos KRBTGT Account: Execute the first of two resets on the Active Directory krbtgt account password to invalidate potentially forged Kerberos Golden Tickets.

4. Restrict Lateral Movement Ports: Enforce host-based firewall policies blocking inbound SMB (TCP 445) and RPC (TCP 135) connections from standard user workstation VLANs to Domain Controllers.

### Phase 1: Eradication & Hardening (Hours 4 – 24) [HIGH / P1]

5. Enable Credential Guard: Enable Windows Defender Credential Guard (Virtualization-Based Security) across all Domain Controllers to isolate LSASS within an isolated virtual enclave and prevent reflective injection.

6. Complete Memory Dump Dissection: Complete Task #8 by executing Volatility 3 malfind and ldrmodules plugins on the memory dump to identify and extract the injected payload.

7. Active Directory Audit: Inspect Active Directory for newly created user accounts, modified security groups, unauthorized shadow administrators, and newly created Group Policy Objects.

8. Second KRBTGT Password Reset: Perform the second reset of the krbtgt account password (after the minimum Kerberos ticket lifetime has expired) to ensure complete invalidation of historical ticket structures.

### Phase 2: Recovery & Monitoring (Hours 24 – 72) [MEDIUM / P2]

9. Deploy Enhanced EDR Telemetry: Install kernel-level Endpoint Detection and Response (EDR) sensors on all Tier-0 and Tier-1 systems with active blocking enabled for LSASS access (PROCESS_VM_READ, PROCESS_VM_WRITE).

10. Restrict Remote Desktop Administration: Mandate Remote Desktop with Restricted Admin mode (/restrictedAdmin) to prevent administrative credentials from being cached in memory on remote target systems.

11. Enforce Protected Users Group: Place all sensitive administrative accounts into the Active Directory 'Protected Users' security group, disabling NTLM authentication, DES/RC4 Kerberos encryption, and unconstrained delegation.

### Phase 3: Extended Forensics & Case Closure (Ongoing) [LOW / P3]

12. Ingest Network Telemetry: Acquire and ingest perimeter proxy logs, DNS query logs, and Windows Security Event Logs into the ForenSight case dossier to resolve identified evidence gaps.

13. YARA & Threat Intelligence Matching: Execute threat intelligence scans on extracted memory buffers to validate or refute the APT39 attribution hypothesis.

14. Final Certified Exhibit Filing: Generate the final certified ForenSight dossier with updated analytical findings for executive legal counsel and regulatory compliance archives.

## 15. Appendices: Evidentiary Manifests & Custody Records

### Appendix A: Digital Evidence Registry & Cryptographic Hashes

| ID | Artifact Name | Type | Size | SHA-256 Digest | Status |
|---|---|---|---|---|---|
| #17 | apt39_memdump.raw | Memory Dump | 56 B | `f1eafcc022fea7e9c7e663ce6919203af72cf96ecbcc693609730cb5d182b3c4` | Registered |

### Appendix B: Chain of Custody & Court Admissibility Trail (2 events)

| Log ID | Timestamp (UTC) | Artifact | Action Taken | Responsible Custodian | Transfer Notes & Verification |
|---|---|---|---|---|---|
| #26 | 2026-10-19 19:23:06 | apt39_memdump.raw | Evidence Ingested | Sarah Connor (Investigator) | Uploaded apt39_memdump.raw (SHA-256: f1eafcc022fea7e9...) |
| #27 | 2026-10-19 19:23:07 | apt39_memdump.raw | Transferred to Secure Vault | Sarah Connor (Investigator) | Evidence secured in Level-3 airgapped evidence repository. |

### Appendix C: Threat Findings Matrix (1 records)

| Finding ID | Target Artifact | Classification | Severity | Recorded Observation Detail |
|---|---|---|---|---|
| #9 | apt39_memdump.raw | Process Injection | Critical | Reflective DLL injection detected inside lsass.exe process address space. |

### Appendix D: Investigation Workflow & Task Execution Log (1 items)

| Task ID | Objective | Assigned Investigators | Due Date | Status |
|---|---|---|---|---|
| #8 | **Analyze Memory Dump for Injected DLLs**<br>Run Volatility3 malfind and ldrmodules plugins on raw memory dump. | Malishka Sachdev, Sarah Connor | 2026-11-01 | Pending |

### Appendix E: Technical Forensic Field Notes (1 notes)

| Note ID | Artifact | Author / Analyst | Timestamp | Field Note Content |
|---|---|---|---|---|
| #7 | apt39_memdump.raw | Sarah Connor (Investigator) | 2026-10-19 19:23:07 | Cryptographic signature matches known APT39 implant variant (Shaheen-IV). |

### Appendix F: Correlated System Audit Activity (6 entries)

| Audit ID | Timestamp (UTC) | Action | Entity Target | Actor | Operational Change Details |
|---|---|---|---|---|---|
| #129 | 2026-10-19 19:23:06 | CREATE | Case #19 | Sarah Connor | Created case via procedure: Operation Nightshade: APT39 Incursion |
| #130 | 2026-10-19 19:23:06 | CUSTODY_LOG | Evidence #17 | Sarah Connor | Custody action logged: Evidence Ingested - Uploaded apt39_memdump.raw (SHA-256: f1eafcc022fea7e9...) |
| #131 | 2026-10-19 19:23:07 | CREATE | Task #8 | Sarah Connor | Created task: Analyze Memory Dump for Injected DLLs |
| #132 | 2026-10-19 19:23:07 | NEW_FINDING | Evidence #17 | System / DB Trigger | Recorded Critical finding: Process Injection - Reflective DLL injection detected inside lsass.exe process address space. |

| | | | | | |
|---|---|---|---|---|---|
| #133 | 2026-10-19 19:23:07 | CUSTODY_LOG | Evidence #17 | Sarah Connor | Custody action logged: Transferred to Secure Vault - Evidence secured in Level-3 airgapped evidence repository. |
| #134 | 2026-10-19 19:23:07 | ADD_NOTE | Evidence #17 | Sarah Connor | Added technical evidence note |

**OFFICIAL FORENSIC CERTIFICATION & DIGITAL SIGN-OFF**

I, **Sarah Connor**, serving as Lead Forensic Analyst, hereby certify under penalty of administrative and legal review that the analytical findings and technical interpretations contained in this ForenSight Incident Dossier (Case #FS-19) accurately represent the observations derived from the secured evidence catalog. All evidence handling adhered to ISO/IEC 27037 standards, with cryptographic SHA-256 integrity maintained throughout custody. Unverified hypotheses have been explicitly labeled.

**Lead Investigator:** Sarah Connor **Role:** Investigator **Date:** 2026-10-20 **Digital Signature:** [SHA256-VERIFIED-19-202610201551]